

Changepoint estimation in climatology

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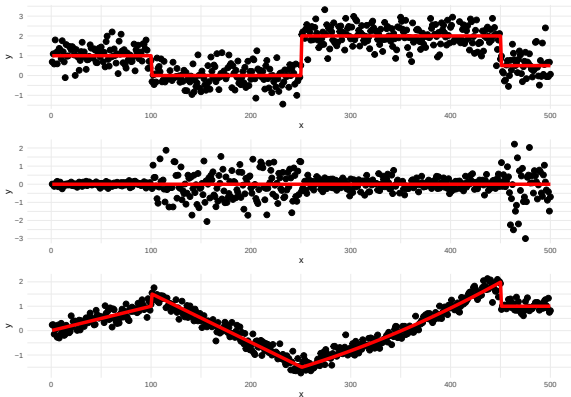
- What are changepoints?
- Fitting multiple changepoint models
- Spatiotemporal changes in air quality
- Exponential decay models for soil moisture

What are changepoints?



For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.



What are changepoints?



Changepoints are also known as:

- breakpoints
- segmentation
- structural breaks
- regime switching
- detecting disorder

and can be found in a wide range of literature including

- quality control
- finance
- medicine
- environment
- linguistics
- ...

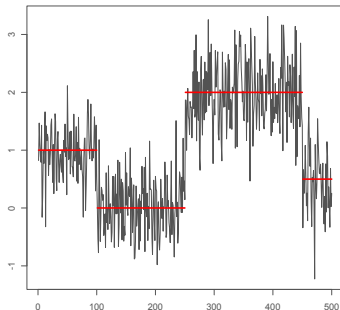
Example



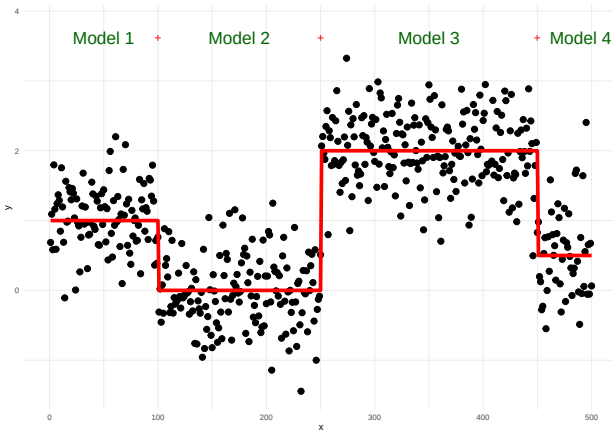
Assume we have time-series data
where

$$Y_t | \theta_t \sim N(\theta_t, 1),$$

but where the means, θ_t , are
piecewise constant through time.



Fitting changepoints



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!



If we define

$$\mathcal{P}_t = \{\tau : 0 < \tau_1 < \dots < \tau_m < t\}$$

$$F(t) = \min_{\tau \in \mathcal{P}_t} \left\{ \sum_{i=1}^{m+1} C(y_{(\tau_{i-1}+1):\tau_i}) + \beta \right\}$$

So,

$$F(t) = \min_{\tau^*} \left\{ \min_{\tau \in \mathcal{P}_{\tau^*}} \left[\sum_{i=1}^m C(y_{(\tau_{i-1}+1):\tau_i}) + \beta \right] + C(y_{(\tau^*+1):t}) + \beta \right\}$$

Thus we minimise,

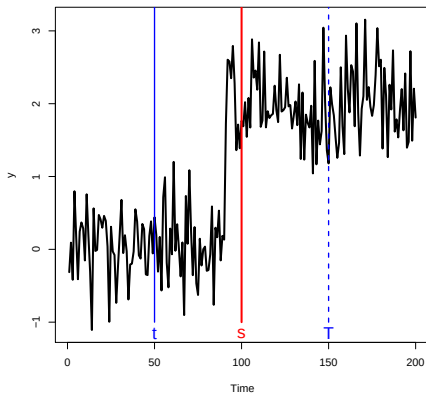
$$F(t) = \min_{\tau^*} \{ F(\tau^*) + \mathcal{C}(y_{(\tau^*+1):t}) + \beta \}$$

Recursively solving the minimisation for $t = 1, \dots, n$ gives an algorithm that is $\mathcal{O}(n^2)$.



The condition in the theorem just means that for any $T > s$ the best partition which involves a changepoint at s will be better than one which have $[t, T]$ as a single segment.

Thus t can never be the (optimal) most recent changepoint prior to T for all $T > s$.



If many t are pruned, excluded from the minimisation then computational time will be drastically reduced.

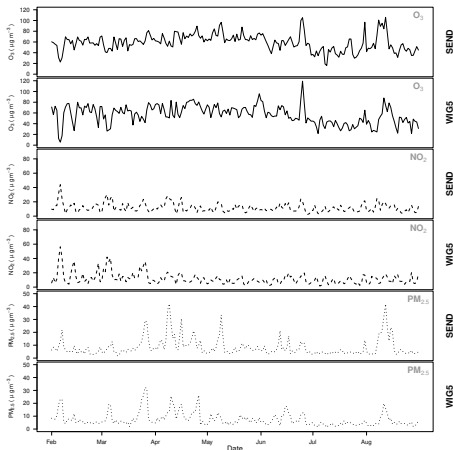
We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.

The most important condition is that *{the number of changepoints increases linearly with n }*.

This is natural in many applications: e.g. as you collect time-series data over larger time-periods; or genomic data over larger regions of the genome.



UK Defra Air Quality Expert group requested scientific evidence around COVID lockdown effects on air quality across the UK.

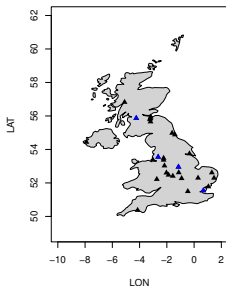




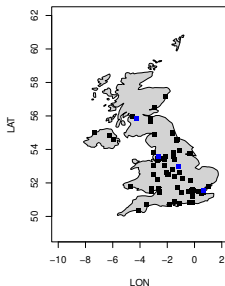
175 AURN active stations, 1st Feb - 31 Aug 2020 (213 days).

O_3 - 30 stations, NO_2 - 74 stations, $PM_{2.5}$ - 30 stations.

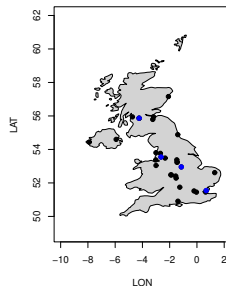
O_3 AURN Sites



NO_2 AURN Sites



$PM_{2.5}$ AURN Sites



OpenAir R package.

Generalised Additive Model

$$y_{s,t} = f_1(x_s) + f_2(x_t) + f_3(x_s, x_t) + \epsilon_{s,t}$$

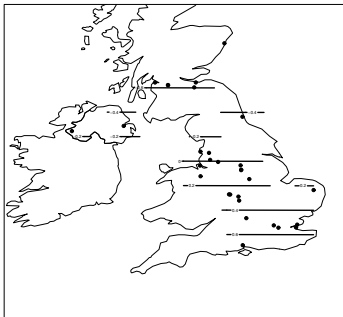
- f_1 is a 2D thin plate spline over space
- f_2 is a cubic regression spline over time
- f_3 is a tensor product for space-time interactions

Use `mgcv` for GAM fit and likelihood.

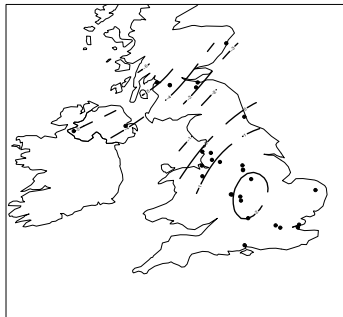


Changepoint at 21st March 2020

PM_{2.5} (pre-lockdown)



PM_{2.5} (lockdown)



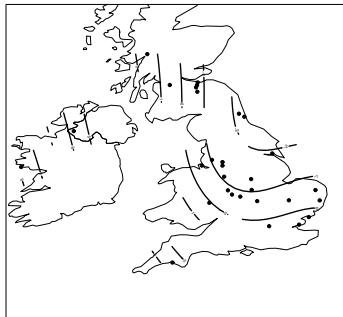


Changepoint at 26th March 2020

O_3 (pre-lockdown)



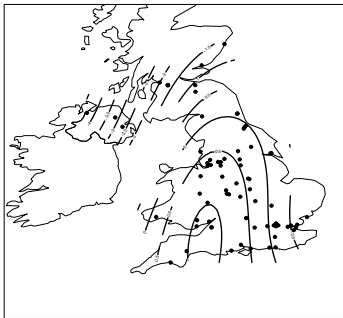
O_3 (lockdown)



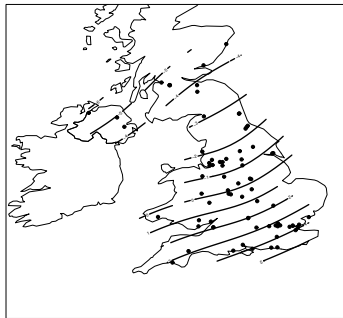


Changepoint at 27th March 2020

NO₂ (pre-lockdown)



NO₂ (lockdown)



- The GAM model is flexible enough to capture spatio-temporal patterns in air quality
- The addition of the PELT step for changepoint detection provides an easy-to-use extension to the model
- Early change for $PM_{2.5}$ correlates with early work from home in cities
- The decrease in O_3 and increase in NO_2 aligns with expectations from science around pollution dissipation
- Changepoints for phased return for schools in June are also seen.

Soil Problem?



- Overall: Monitor soil health
- Soil moisture: Important component
- Understand how soil moisture dynamics change over time
- Understand how soil moisture dynamics relate to soil health

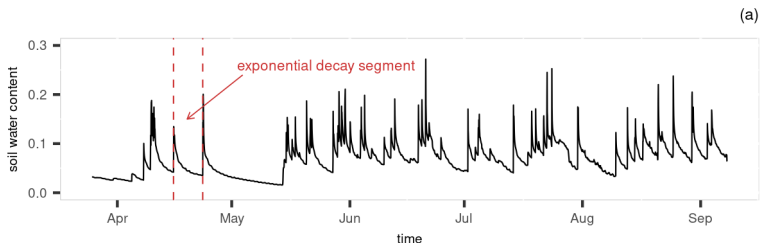
Problems:

- Long term (open source) data is scarce (NEON)
- Documented changes in soil health don't exist
- Existing approaches require manual intervention





Soil water content can **increase suddenly** following a weather event (e.g. a storm). Then it will **decrease towards an asymptotic level**, until the next event disrupts the process and leads to another sudden increase.

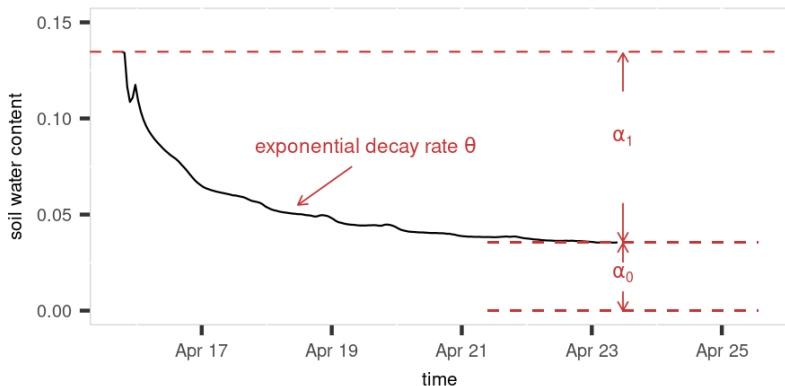




In particular, the segment following a sudden increase is assumed to follow an **exponential decay process**, which can be modeled as

$$Y_t = \alpha_0 + \alpha_1 \theta^{(t-\tau)} + \epsilon_t$$

(b)



- The **time when the sudden increase occurs** is considered as a changepoint.
- The segment following the changepoint is described using an **exponential decay model**, parameterised as

$$Y_t = \alpha_0 + \alpha_1 \theta^{(t-\tau)} + \epsilon_t = \alpha_0 + \alpha_1 \exp[-\exp(\gamma) (t - \tau)] + \epsilon_t .$$

- The parameters in the exponential decay model are assumed to **vary from segment to segment**. The varying parameters have the potential of capturing the temporal dynamics unique to the soil under investigation.

- Motivated by the concept of a hydrological signature in hydrological literature, where **physically interpretable hydrological indices** are defined and **extracted from observable continuous data**, we made our attempt to establish insights into meaningful **soil properties** from time series data.
- We applied the **changepoint-based method** to the **soil water content time series** from the NEON data portal (<https://data.neonscience.org/>) and extracted information from the model to create **soil signatures**.

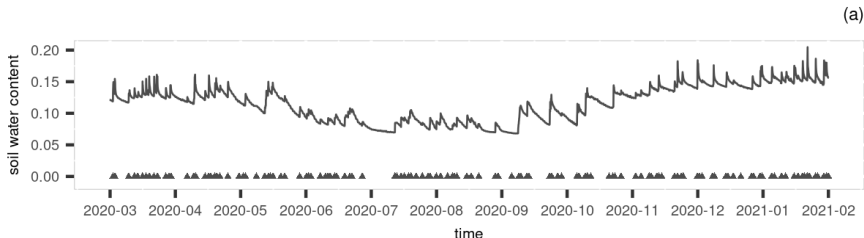
NEON soil moisture

Nine NEON field sites with hourly time series data of at least nine months.



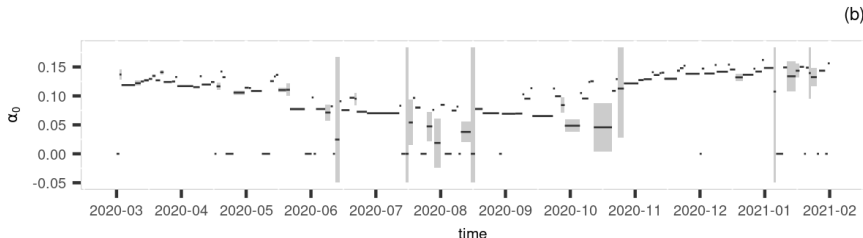


Soil volumetric water content time series from NEON site ORNL. The black triangles indicate the estimated changepoint.



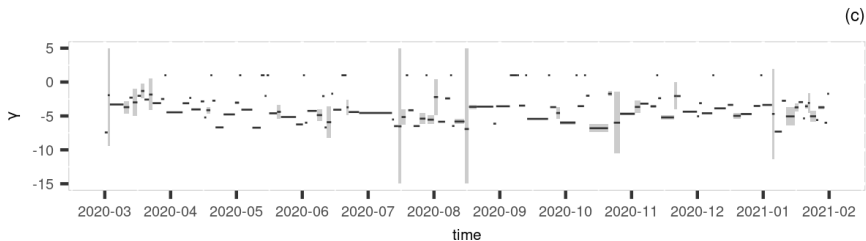


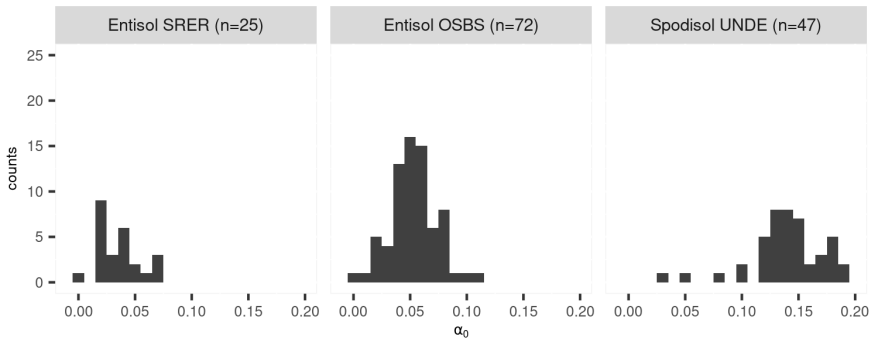
The estimated asymptotic parameter α_0 plotted over time, along with the 95% confidence intervals plotted as the grey bands.

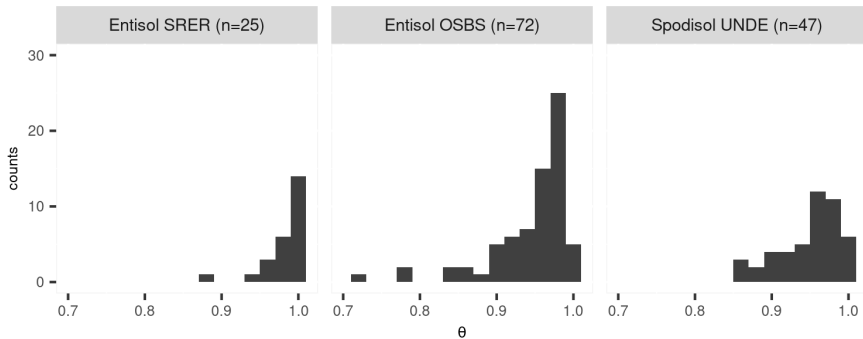




The estimated decay parameter plotted over time in the scale used during the optimisation, i.e. $\gamma = \log(-\log(\theta))$, $\gamma \in \mathcal{R}$, along with the 95% confidence intervals plotted as the grey bands.







Correlation to commonly used soil properties

	$\hat{\alpha}_0$			$\hat{\theta}$		
	1st qt	median	3rd qt	1st qt	median	3rd qt
MAT (deg)	-0.6603*	-0.6996*	-0.8016***	0.1556	0.1698	0.0713
MAP (mm)	0.2642	0.3336	0.1889	-0.6267*	-0.6423*	-0.5486
sand total	-0.0704	-0.1770	-0.0985	-0.2377	-0.3324	-0.3423
silt total	0.2542	0.3517	0.2746	0.1722	0.2522	0.2704
clay total	-0.2558	-0.1452	-0.2173	0.3249	0.4346	0.4290

Note: significance level from a Pearson correlation test is indicated using stars where *** (≤ 0.01), ** (≤ 0.05), * (≤ 0.1).

- The **distributions** shown by the histograms of the estimated asymptotic and decay parameters **varies across sites**.
- The differences can be attributed to the **climate** of the field sites (e.g. annual precipitation, mean temperature) and their **soil compositions** (i.e. sand, silt and clay content).
- The **distributions, along with their summary statistics**, may be regarded as the **signatures** of a particular soil.
- In soils that experience changes (e.g. degradation, restoration), **changes of the signatures** may be expected.

- Introduced the PELT algorithm for identifying multiple changepoints optimally in a computationally efficient way
- Demonstrated versatility of the approach with different models including:
 - Developed an approach to identify changepoints in spatiotemporal data with an application in air quality
 - Developed an approach to automatically quantify changes in soil moisture dynamics using exponential decay segments
- These and other extensions are useful for identify changes in a host of climatology applications